

Reflection: Multi-Agent Workflow Planner for a Startup Accelerator

VT_AGI: Dev Tools & Product Readiness Module Project

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2026-07-20

1. Project Overview

This project implements a multi-agent workflow planner for a startup accelerator. A founder supplies a single input -- their startup's domain, e.g. "fintech" -- and three collaborating agents produce a founder-ready pitch deck outline: a Research Agent gathers domain-specific market trends, a Funding Advisor recommends grants and funding programs suited to that domain, and a Pitch Coach synthesizes both into a seven-section outline and decides whether the result is complete or needs another pass. The system runs through two interfaces (a CLI and a Jupyter/ipywidgets notebook GUI) that share one workflow implementation, persists prior findings in a local vector store so agents are not starting from nothing on repeated runs, and logs every agent transition both locally and, optionally, to Azure Application Insights.

2. Architecture

The workflow is a LangGraph StateGraph over a single shared, typed state object. The two diagrams below present the same system from two angles -- an end-to-end workflow diagram of the sequence including the conditional feedback loop, and a swimlane diagram that makes clear which element (agent) is responsible for which action.

Figure A -- End-to-End Workflow Diagram

The full pipeline from domain input through Research Agent, Funding Advisor, and Pitch Coach, to the completion check. Every agent transition is logged locally as JSONL, and to Azure Application Insights, which is configured for this project (see Stage 3 in Section 8). If the Pitch Coach's verdict is not COMPLETE, the graph routes back to the Research Agent (NEEDS_MORE_RESEARCH) or the Funding Advisor (NEEDS_MORE_FUNDING) with feedback_notes set, capped at MAX_PITCH_REVISIONS = 2 so the loop always terminates regardless of model behavior.

Multi-Agent Pitch Planner -- End-to-End Workflow

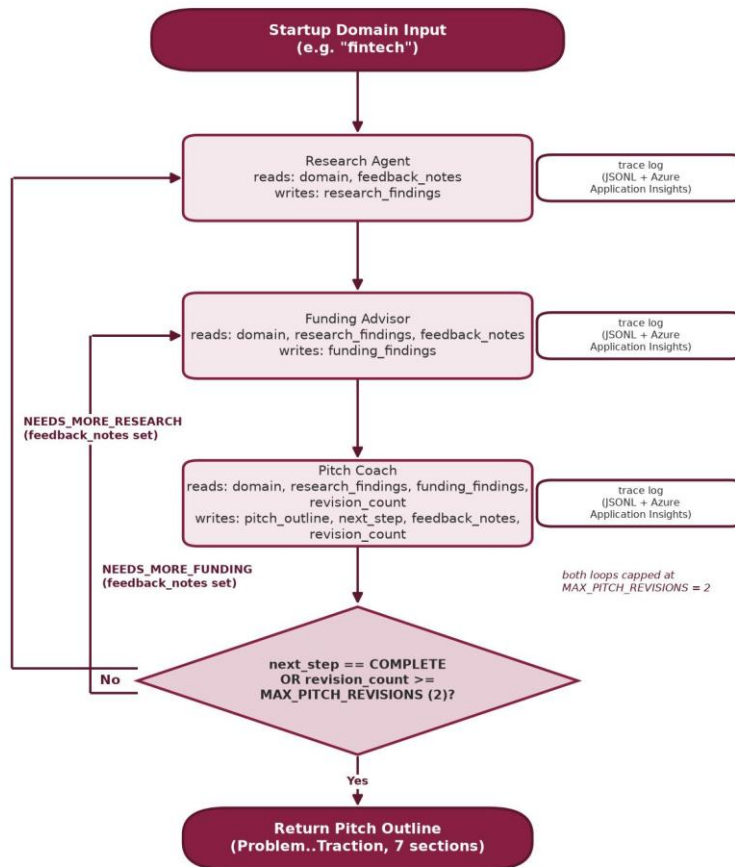
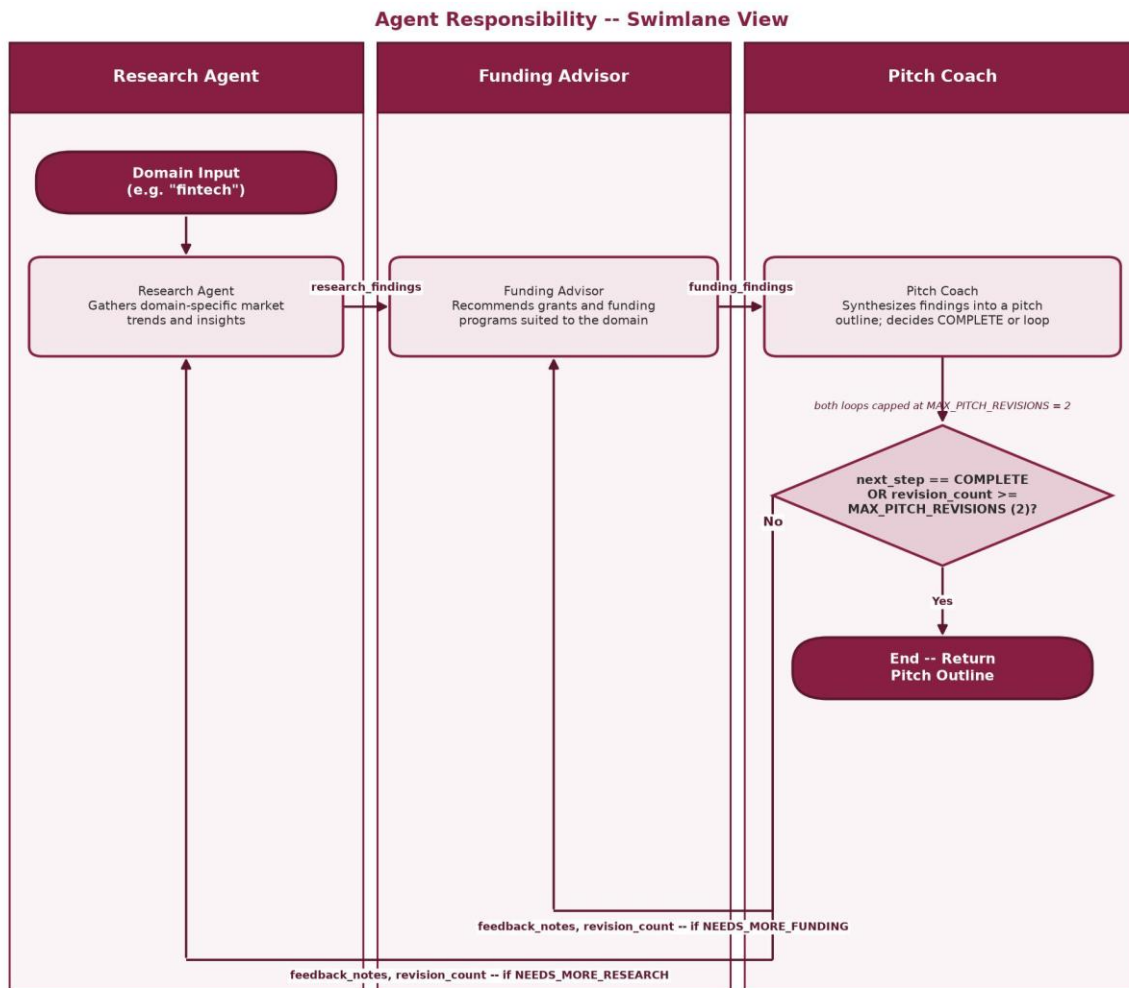


Figure B -- Agent Responsibility Swimlane

The same workflow viewed as a swimlane: one lane per agent, showing which element is responsible for which action and how findings and feedback cross lane boundaries. Responsibilities are scoped with no overlap -- only the Research Agent produces market findings, only the Funding Advisor produces funding recommendations, and only the Pitch Coach writes the final outline or triggers a feedback loop back into an earlier lane.



3. Design Decisions

Orchestration framework -- LangGraph over CrewAI/AutoGen

The rubric required justifying one of CrewAI, LangGraph, or AutoGen based on task complexity and interaction patterns, but none of the three frameworks was named anywhere in the course's own support materials, so the choice was independently justified rather than pattern-matched to something taught. LangGraph was chosen because the rubric explicitly requires a feedback-driven flow and reasoning traceability with memory/context handoffs -- LangGraph's state graph makes both first-class: cyclic/conditional edges implement the feedback loop natively rather than as glue code, and its explicit typed state object is inspectable at every node transition, which is exactly what the logging requirement needs. CrewAI maps roles to Agents more literally (arguably a more intuitive fit for Research Agent / Funding Advisor / Pitch Coach) but has weaker native support for cyclic feedback and state-level traceability. AutoGen's group-chat pattern is a poor fit for a fixed, three-role pipeline with non-overlapping responsibilities.

LLM backend -- OpenAI

Course Demo_02 taught Ollama for local/offline LLM use, but only as a privacy/offline alternative, not a requirement -- the course's own Guided Practice used a hosted API (Azure OpenAI) as the default. The project uses the OpenAI API exclusively (gpt-4.1-mini by default, via langchain_openai.ChatOpenAI, configured once in

`config.get_llm()` and reused by all three agents): existing OpenAI account access and credits were already available, making it the path of least friction for a hosted-API backend with no local model path required.

Vector memory -- local Chroma over a managed vector DB

No vector memory tool was taught anywhere in the course materials, so this choice was also independent. Local Chroma was chosen for zero external-service setup while still giving genuine semantic retrieval of prior agent findings, over a heavier option like a managed cloud vector DB that would have added an account/credential dependency for no clear benefit at this project's scale.

Interface -- CLI + Jupyter/ipywidgets notebook, not Colab

The course's own demonstrated interface pattern is a Jupyter notebook with ipywidgets (Textarea/Button/Output), run in Colab. This project keeps that same widget pattern but runs it locally in VS Code's Jupyter extension instead of Colab, plus adds a CLI as the second, simpler entry point -- both call the same underlying `workflow.graph.run_workflow()` so there is exactly one implementation, not two.

Observability -- Azure Application Insights, scoped deliberately

This is a Microsoft-sponsored course, which created a pull toward "use Azure somewhere," but the LLM calls run through a plain OpenAI API key, not an Azure-hosted endpoint -- there was no natural place to put Azure OpenAI without displacing a decision already made for good reasons. Instead, Azure was scoped specifically to observability: the course's own Instructor Slides deck is literally titled "Agent Deployment Readiness and Scaling Strategies" and covers observability, monitoring, and scaling directly, so Application Insights ties to real taught course content rather than being Azure-for-Azure's-sake. It is optional and gracefully degrades to local-only logging if not configured, since it was not certain at build time whether an Azure subscription would even be available -- confirmed correct in practice once Stage 3 walked through creating that subscription from scratch.

4. Challenges and Resolutions

Feedback-loop termination

A cyclic graph needs a guaranteed exit condition or it can loop indefinitely. `MAX_PITCH_REVISIONS` caps the loop at a small fixed number of passes regardless of what the model outputs, so termination does not depend on the model behaving well.

Routing signal reliability

The Pitch Coach's routing decision is parsed from a single verdict line the model is asked to append, rather than a second structured-output call. This keeps the design simple (one call per agent) at the cost of being slightly less robust than a dedicated structured-output schema; if the model omits the verdict line, the code defaults to ending the workflow rather than looping or erroring.

First Azure login changed the screenshot plan

The user had never signed into Azure before, so account creation (identity verification, free-trial signup) turned out to be its own step ahead of subscription confirmation, not something bundled into it. The Stage 5 screenshot inventory was revised mid-walkthrough to insert that step and renumber the rest -- a small example of a plan needing to flex once real hands-on execution started.

Azure Portal UI has drifted from older documentation

The original guidance assumed a classic-vs-workspace-based radio button when creating an Application Insights resource. The current Portal UI has removed that choice -- it defaults straight to workspace-based, signaled only by

a "Workspace Details" section that auto-selects a new Log Analytics Workspace. Similarly, the connection string only offers a copy-to-clipboard icon rather than an inline expandable field, which was worked around by pasting the clipboard contents into a scratch text editor just to view and screenshot the full value.

5. Trade-offs

The following table summarizes the key trade-offs made during the build:

Decision	Trade-off
LangGraph vs. CrewAI/AutoGen	First-class cyclic feedback and inspectable state; steeper conceptual model than CrewAI's more literal role-to-Agent mapping
Single LLM provider (OpenAI)	Simple, single code path, one credential to manage; no automatic fallback if OpenAI is unavailable
Local Chroma vs. managed vector DB	Zero external-service setup, no added credential dependency; not horizontally scalable or cloud-native
CLI + local notebook vs. Colab	Consistent local dev experience, no framework lock-in; requires local Python/Jupyter setup instead of zero-install browser access
Azure Application Insights optional	Runs safely with no cloud account at all; only adds real value once actually configured, which depended on Stage 3 succeeding
Feedback loop re-runs both upstream agents	One conditional edge, simple graph; an extra Funding Advisor call on a research-only revision, since fixed edges always flow research -> funding -> pitch coach

6. Conclusion

The project delivers a working end-to-end multi-agent pitch planner, verified with four live-LLM runs across three domains (fintech, urban vertical farming, telehealth for elder care) through both interfaces. Local trace logging and optional Azure Application Insights telemetry both work correctly, confirmed by matching custom span properties against the local JSONL logs rather than assuming configuration alone was sufficient. The one honest gap: none of the four live runs happened to trigger the feedback loop, so MAX_PITCH_REVISIONS and the revision-routing logic remain verified only through the mocked test suite, not a live-LLM revision pass -- a reasonable limitation to name rather than paper over, and a natural next step if this project continued.

7. Personal Reflections

On the decision process itself

Rather than guessing at requirements and risking an expensive-to-unwind wrong architectural assumption, the framework/backend/interface/memory/Azure decisions were each surfaced as explicit questions before any code was written. This front-loaded some back-and-forth but meant the build stage proceeded without having to revisit any of those five decisions mid-build.

On reconciling course branding with real constraints

A Microsoft/VT-branded course naturally invites "make it Azure-native," but forcing that everywhere (e.g. Azure OpenAI as the LLM backend) would have fought against the simpler, already-justified choice of a plain hosted API for the LLM calls. Scoping Azure to the one place -- observability -- where it adds genuine, course-aligned value is itself a small case study in not letting a platform affiliation dictate every technical choice, even though Azure OpenAI would have been a technically viable alternative to the plain OpenAI API key used here.

On the hands-on stages

This is meant as honest feedback, not a complaint: the friction in Stages 3 and 4 was rarely in the actual Azure or CLI work itself, and much more often in small tooling mismatches -- VS Code's Python interpreter picker not surfacing the project's own local .venv, ipywidgets output silently not surviving a notebook save, and more than one suggested filename using a colon that Windows immediately rejected. None of these were hard problems once identified, but they consumed more back-and-forth than the underlying task did, which was a genuinely useful reminder that environment friction is real project work, not a distraction from it. The colon-in-filename mistake in particular happened more than once before it was corrected for good, and a screenshot that was assumed captured turned out not to exist until the gap was caught later -- the pattern that worked best afterward was treating every claimed "done" as something to verify, not something to trust by default.

8. Build Walkthrough -- Screenshots

The following screenshots document each stage of the build in sequence, from environment verification through Azure configuration to live end-to-end verification. Each image corresponds to a checklist item in the project ROADMAP's Screenshot Inventory.

Figure 1 -- Python 3.12 and Virtual Environment Confirmed

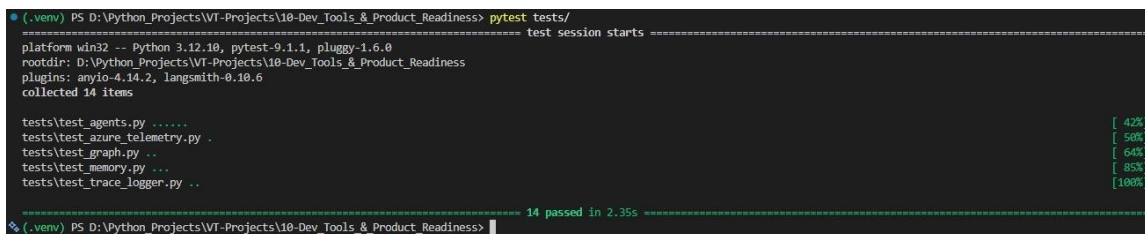
PowerShell with the project's .venv activated and python --version reporting Python 3.12.10, matching the CLAUDE.md baseline. Captured retroactively after Stage 2's build had already finished, since re-running the same commands reproduces this state identically.



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE
(.venv) PS D:\Python_Projects\VT-Projects\10-Dev_Tools_&_Product_Readiness> .venv\Scripts\activate
(.venv) PS D:\Python_Projects\VT-Projects\10-Dev_Tools_&_Product_Readiness> python --version
Python 3.12.10
(.venv) PS D:\Python_Projects\VT-Projects\10-Dev_Tools_&_Product_Readiness>
```

Figure 2 -- Full pytest Suite Passing

All 14 tests across five test modules (agents, graph, memory, Azure telemetry graceful degradation, trace logger) pass in 2.35 seconds against mocked OpenAI responses -- no API key or network access required.



```
(.venv) PS D:\Python_Projects\VT-Projects\10-Dev_Tools_&_Product_Readiness> pytest tests/
===== test session starts =====
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.6.0
rootdir: D:\Python_Projects\VT-Projects\10-Dev_Tools_&_Product_Readiness
plugins: anyio-4.14.2, langsmith-0.10.6
collected 14 items

tests\test_agents.py ..... [ 42%]
tests\test_azure_telemetry.py . [ 50%]
tests\test_graph.py .. [ 64%]
tests\test_memory.py ... [ 85%]
tests\test_trace_logger.py .. [100%]

===== 14 passed in 2.35s =====
(.venv) PS D:\Python_Projects\VT-Projects\10-Dev_Tools_&_Product_Readiness>
```

Figure 3 -- Azure Account Creation

The Azure Portal home screen immediately after first-ever sign-in, showing no existing subscription and the "Start with an Azure free trial" option. Account creation turned out to be its own step ahead of subscription confirmation, since this was the user's first time logging into Azure.

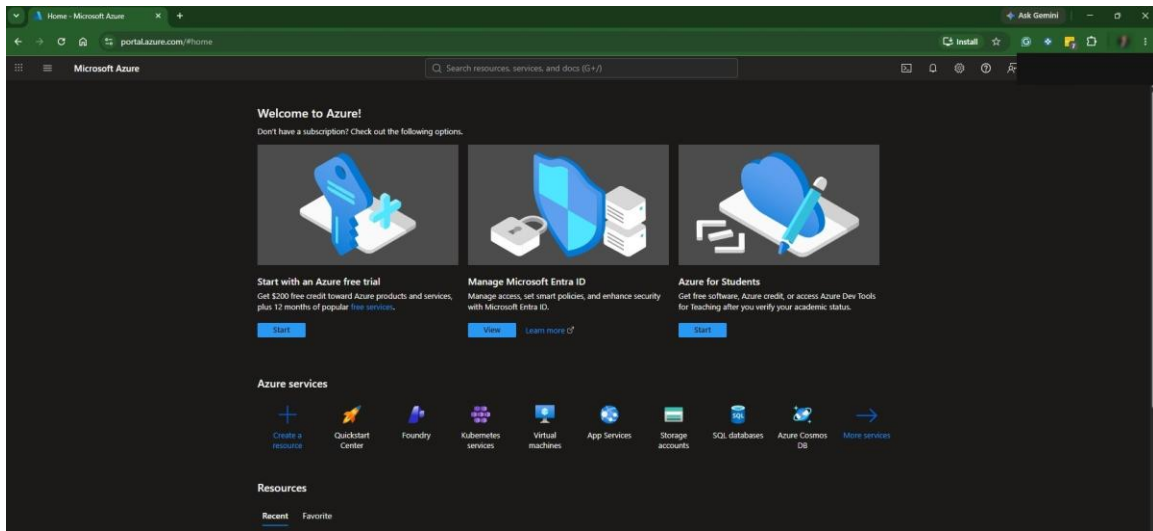


Figure 4 -- Subscription Confirmed

The Subscriptions blade showing "Azure subscription 1" active with Owner role, confirming the free trial signup succeeded before moving on to resource creation.

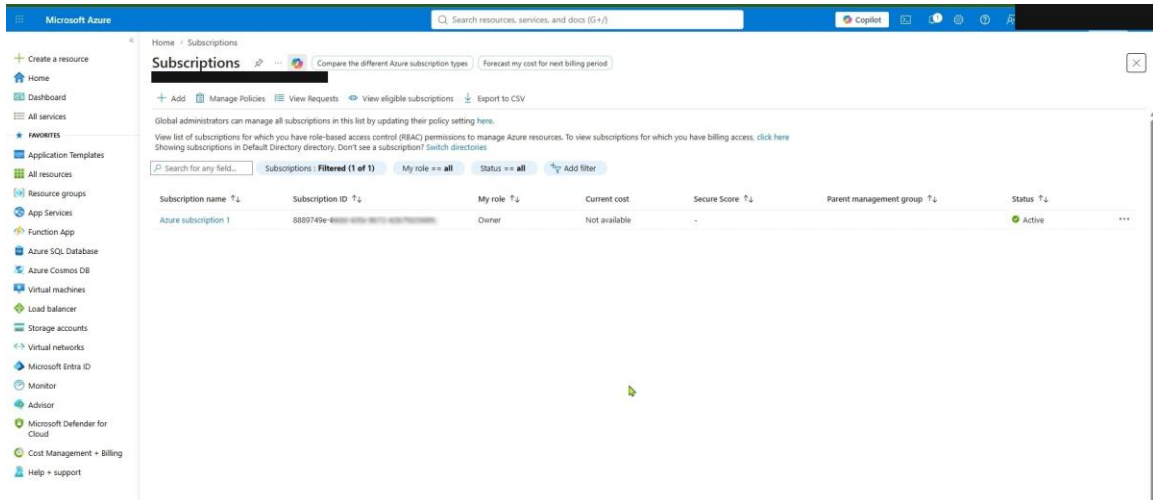


Figure 5 -- Resource Group Basics

The Basics tab of the "Create a resource group" form: subscription "Azure subscription 1", name vt-agi-10-accelerator-agents, region East US.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ Azure subscription 1

Resource group name * ⓘ vt-agi-10-accelerator-agents

Region * ⓘ (US) East US

Previous Next Review + create

Figure 6 -- Resource Group Review + Create

The Review + create summary confirming the resource group settings before submission.

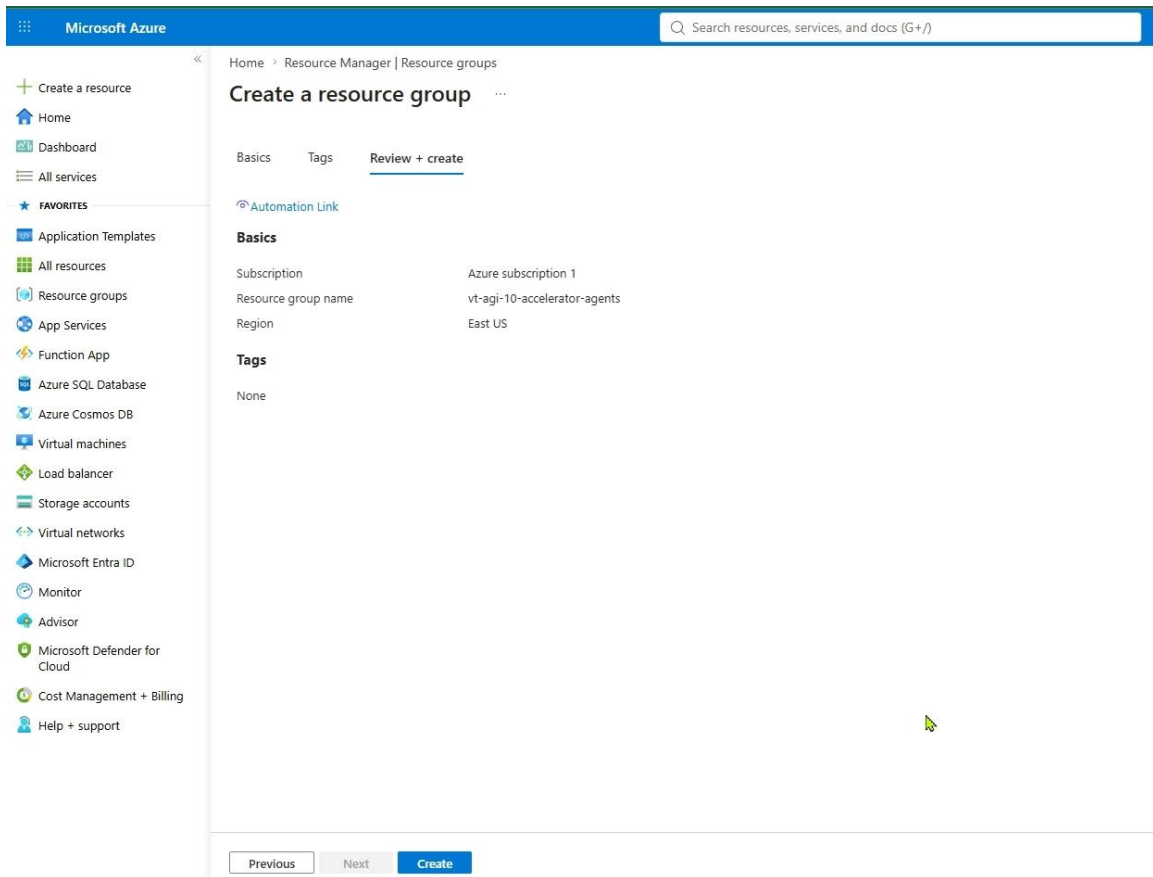


Figure 7 -- Resource Group Created

The Resource groups list confirming vt-agi-10-accelerator-agents now exists under Azure subscription 1 in East US.

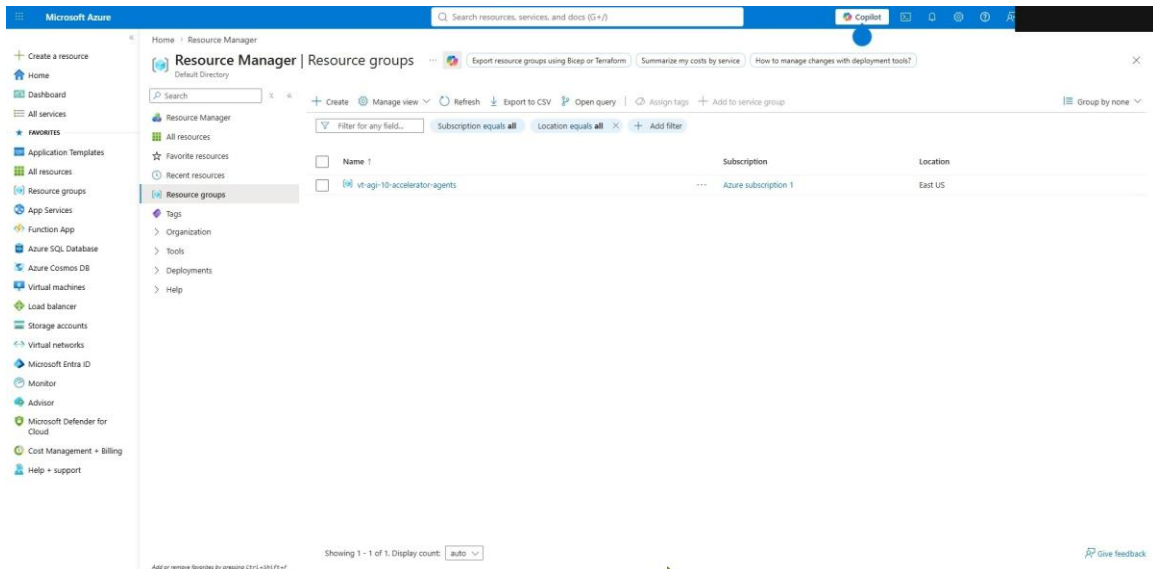


Figure 8 -- Application Insights Resource Created

The Overview page of the newly created appi-accelerator-agents Application Insights resource, workspace-based, with the connection string and instrumentation key visible in the Essentials panel. The current Azure Portal no longer shows a classic-vs-workspace-based toggle during creation -- it defaults straight to workspace-based.

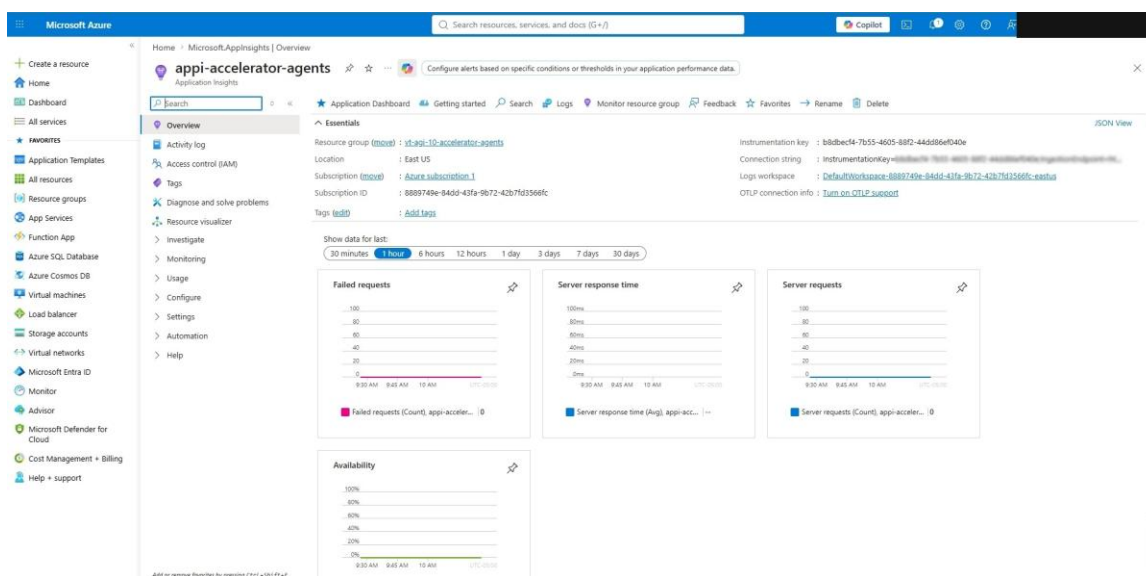


Figure 10 -- Connection String Added to .env

AZURE_APPINSIGHTS_CONNECTION_STRING set in the local .env file (values redacted), with the comment corrected from "Required" to "Optional -- Azure Application Insights" to stay consistent with the code's graceful degradation.

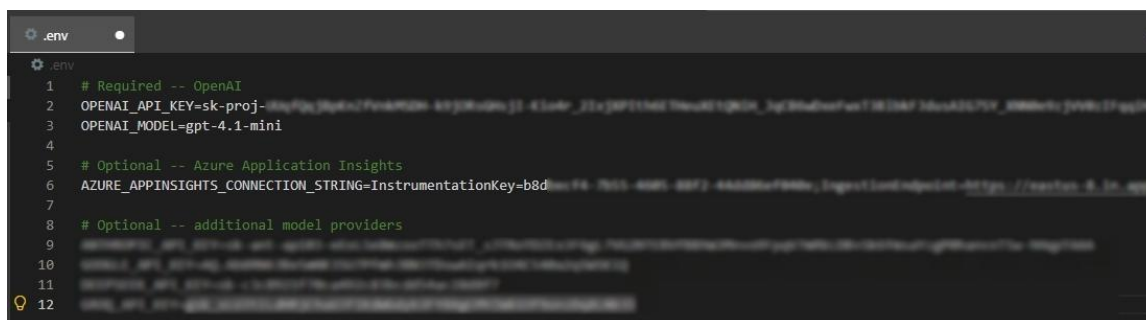


Figure 11 -- CLI Run: fintech (Terminal Output)

python -m interface.cli --domain fintech producing a full seven-section pitch outline and confirming the trace log file path. This run (and a second fintech run captured separately, Figure 12) completed in a single pass with revision_count: 0, next_step: "end" -- no feedback loop fired.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE
(.venv) PS D:\Python_Projects\VT-Projects\10-Dev_Tools_& Product_Readiness> .venv\Scripts\python -m interface.cli --domain fintech

2. **Solution**
- Develop an AI-powered credit scoring model leveraging alternative data to improve accuracy and broaden access to underserved customers.
- Integrate DeFi protocols with traditional banking APIs to create trusted hybrid financial products offering the best of both worlds.
- Build an embedded finance platform enabling seamless in-app payment, lending, and insurance offerings across multiple fintech ecosystems.
- Implement transparent, regulatory-compliant BNPL services with flexible repayment terms prioritizing consumer protection.
- Deploy advanced biometric authentication and real-time fraud detection systems to enhance security and reduce transaction risk.

3. **Market**
- Target underserved and underbanked customer segments excluded by traditional credit models globally.
- Fintech users seeking hybrid financial products that combine DeFi innovations with trusted banking services.
- App developers and fintech platforms in need of embedded finance solutions to expand service offerings and user engagement.
- BNPL market participants facing increasing regulatory scrutiny and demand for transparent, flexible credit options.
- Digital financial service providers prioritizing advanced security technologies to safeguard transactions.

4. **Business Model**
- Licensing AI-powered credit scoring technology to financial institutions and lending platforms on a subscription or usage basis.
- Charging transaction fees and integration licensing for hybrid DeFi and traditional banking APIs usage.
- Revenue share or SaaS fees from embedded finance platform integration partners across payments, lending, and insurance verticals.
- Service fees from BNPL offerings with transparent pricing and compliance support to merchants and consumers.
- Security-as-a-service model providing biometric and fraud detection solutions to digital finance platforms.

5. **Funding Ask**
- Seeking $X million to finalize AI-credit scoring model development and DeFi-traditional banking API integrations.
- Funds will also support product enhancements in embedded finance features, BNPL regulatory compliance capabilities, and security systems.
- Allocation towards obtaining necessary certifications and pilot partnerships with financial institutions and app ecosystems.

6. **Team**
- Founders with expertise in AI, fintech product development, blockchain/DeFi integration, and regulatory compliance.
- Advisors include fintech security experts and former banking regulators experienced in BNPL and digital finance policies.
- Key hires planned in data science, blockchain development, and compliance/legal affairs.

7. **Traction**
- Early prototype of AI-powered credit scoring model tested on alternative data sets showing improved risk accuracy.
- MVP integrating select DeFi protocols with banking APIs under development with ongoing pilot discussions.
- Initial embedded finance partnerships explored with fintech app developers.
- BNPL service framework aligned with emerging regulatory guidelines established.
- Security solution pilot completed with positive fraud detection results in controlled environments.

[INFO] Trace log written to D:\Python_Projects\VT-Projects\10-Dev_Tools_& Product_Readiness\logs\run_20260720T163431-0f3003c5.jsonl
(.venv) PS D:\Python_Projects\VT-Projects\10-Dev_Tools_& Product_Readiness> []
```

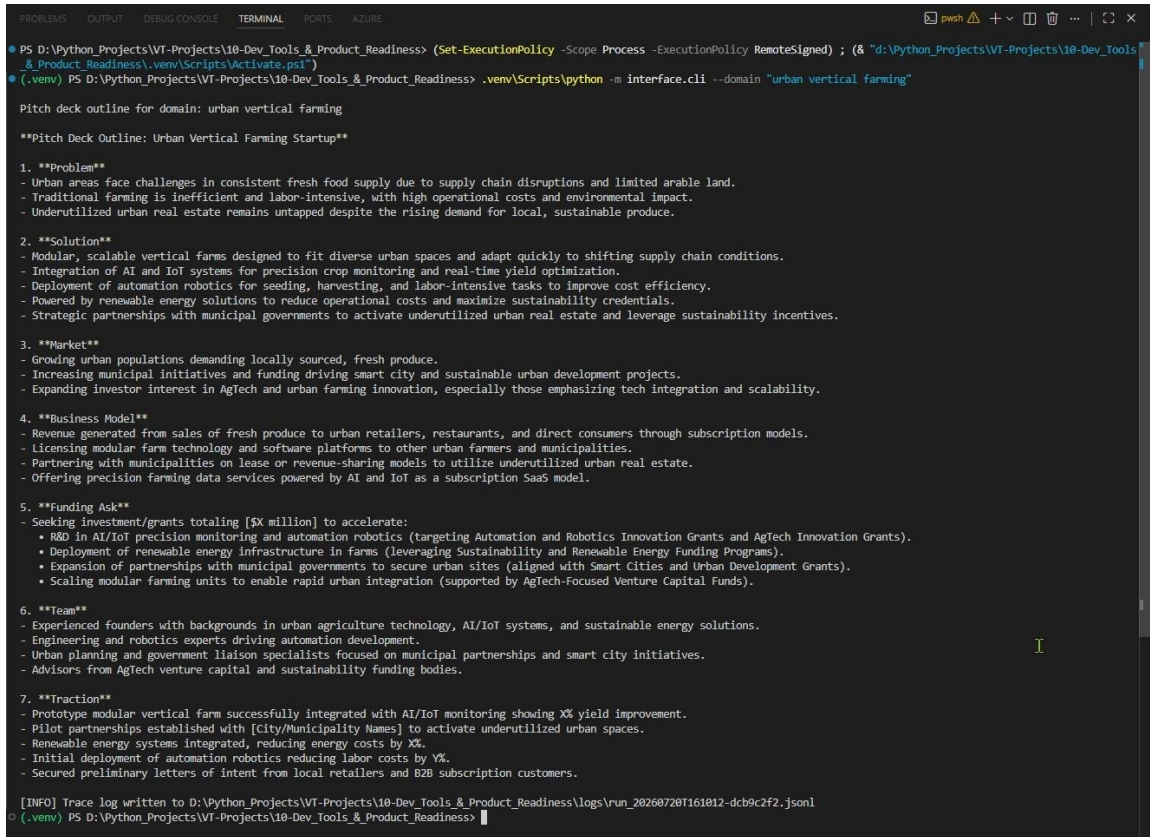
Figure 12 -- CLI Run: fintech (Trace Log)

The JSONL trace log for the original fintech run, opened in VS Code, showing one entry per agent (research_agent, funding_advisor, pitch_coach) with the domain, revision count, and final verdict recorded.

```
run_20260720T155421-fcf7bd36.jso
File Edit View
{"timestamp": "2026-07-20T15:54:27.760580+00:00", "run_id": "20260720T155421-fcf7bd36", "agent": "research_agent", "revision_count": 0, "state_read": {"domain": "fintech", "feedback_notes": ""}, "state_written": {"research_findings": "- Increasing adoption of AI-driven credit scoring and risk assessment to expand financial inclusion. \n- Emergence of decentralized finance (DeFi) protocols integrating with traditional banking services. \n- Growing demand for embedded finance solutions enabling seamless payment and lending within non-financial apps. \n- Expansion of Buy Now, Pay Later (BNPL) models with regulatory scrutiny driving innovation in transparent credit offerings. \n- Heightened focus on cybersecurity and fraud prevention leveraging biometrics and real-time transaction monitoring."}}
{"timestamp": "2026-07-20T15:54:31.498911+00:00", "run_id": "20260720T155421-fcf7bd36", "agent": "funding_advisor", "revision_count": 0, "state_read": {"domain": "fintech", "feedback_notes": ""}, "state_written": {"funding_findings": "- **Fintech and AI Innovation Grants** \n Tailored for startups applying AI in financial services, these grants support development of AI-driven credit scoring and risk assessment tools, aligning with the trend toward expanding financial inclusion.\n\n- **Government Digital Finance and Inclusion Funding Programs** \n Focused on promoting digital financial services and inclusion, these programs back initiatives integrating DeFi with traditional banking and embedded finance solutions to broaden access and user reach.\n\n- **Cybersecurity and Fraud Prevention Grants** \n Offered by public agencies or industry bodies targeting advancements in biometric security and real-time transaction monitoring, addressing the critical need for enhanced cybersecurity in fintech.\n\n- **Regulatory Innovation and Compliance Support Grants** \n Designed to help startups navigate increasing regulatory scrutiny (e.g., in BNPL), supporting transparent and compliant credit offerings through innovation and collaboration with regulators.\n\n- **Sector-Focused Venture Capital Funds Specializing in Fintech** \n VC funds with fintech mandates often prioritize startups driving embedded finance, DeFi integration, and AI-powered financial products, providing not only capital but domain expertise aligned with market trends."}}
{"timestamp": "2026-07-20T15:54:39.656215+00:00", "run_id": "20260720T155421-fcf7bd36", "agent": "pitch_coach", "revision_count": 0, "state_read": {"domain": "fintech", "feedback_notes": ""}, "state_written": {"pitch_outline": "***Pitch Deck Outline**\n\n1. **Problem** \n- Millions remain excluded from traditional financial services due to outdated credit scoring methods and lack of transparent, accessible credit options. \n- Integration challenges between decentralized finance (DeFi) protocols and traditional banking hinder broader adoption and user experience. \n- Non-financial apps lack embedded finance capabilities, limiting seamless payment, lending, and credit services within ecosystems customers frequently use. \n- Buy Now, Pay Later (BNPL) services face rising regulatory scrutiny demanding transparent, compliant credit products. \n- Increasing cyber threats and fraud in financial transactions threaten user trust and platform security.\n\n2. **Solution** \n- AI-driven credit scoring and risk assessment platform designed to expand financial inclusion with more accurate, unbiased credit evaluation. \n- Bridging technology enabling smooth integration of DeFi protocols with legacy banking infrastructures to create hybrid financial offerings. \n- Embedded finance toolkit allowing non-financial applications to seamlessly offer payment, lending, and credit services within their user experience. \n- Transparent and regulatory-compliant BNPL product features developed in partnership with regulators to ensure trust and innovation. \n- Advanced cybersecurity framework leveraging biometrics and real-time transaction monitoring to detect and prevent fraud robustly.\n\n3. **Market** \n- Large and rapidly growing fintech market driven by AI adoption, digital financial inclusion, and regulatory demand for innovation in credit products. \n- Target segments include underserved consumers and SMEs seeking access to credit, non-financial platforms aiming to embed finance, and regulated BNPL providers adapting to compliance requirements. \n- Opportunity spans emerging and developed markets where AI-driven risk tools and DeFi integrations are gaining traction alongside embedded finance growth.\n\n4. **Business Model** \n- SaaS subscription fees for AI credit scoring and risk assessment services licensed to banks, fintechs, and lenders. \n- API-based revenue-sharing or licensing for embedded finance solutions deployed within third-party apps. \n- Transaction fees or interest margin splits on BNPL products offered through partner platforms. \n- Consulting and compliance support services to assist clients in navigating regulatory frameworks, especially for BNPL offerings. \n- Recurring subscription or licensing fees for cybersecurity tools focused on biometrics and real-time monitoring.\n\n5. **Funding Ask** \n- Seeking $X million to accelerate AI model development and integration capabilities for DeFi and embedded finance. \n- Funds allocated for expanding cybersecurity technology with advanced biometric systems and real-time fraud detection. \n- Capital to support regulatory collaboration initiatives ensuring compliant, transparent BNPL rollout. \n- Investment to build partnerships with financial institutions and non-financial app platforms to scale adoption.\n\n6. **Team** \n- Founders with deep fintech expertise in AI-driven risk assessment, regulatory compliance, and embedded finance. \n- Experienced technologists proficient in DeFi protocols, API development, and biometric security systems. \n- Advisors with backgrounds in banking regulation, cybersecurity, and venture funding. \n- Cross-functional team combining product development, legal/regulatory affairs, and business development capabilities.\n\n7. **Traction** \n- Successfully piloted AI credit scoring with early lending partners, demonstrating improved default prediction accuracy. \n- Proof-of-concept integration completed connecting DeFi liquidity pools with traditional banking APIs. \n- Embedded finance modules currently in beta with select non-financial apps showing increased user engagement metrics. \n- Positive regulatory feedback on BNPL compliance framework under development. \n- Initial deployment of biometric fraud prevention system showing reduced transaction fraud rates in live tests.", "next_step": "end", "feedback_notes": ""}}
```


Figure 13 -- CLI Run: urban vertical farming (Terminal Output)

A second domain, chosen specifically to be narrower and less common than fintech in an attempt to trigger a Pitch Coach revision verdict. It also completed in a single pass -- still a valid domain-generality data point even without the loop firing.



```
PS D:\Python_Projects\VT-Projects\10-Dev_Tools_& Product_Readiness> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "d:\Python_Projects\VT-Projects\10-Dev_Tools_& Product_Readiness\venv\Scripts\Activate.ps1")
(.venv) PS D:\Python_Projects\VT-Projects\10-Dev_Tools_& Product_Readiness> .venv\Scripts\python -m interface.cli --domain "urban vertical farming"

Pitch deck outline for domain: urban vertical farming

**Pitch Deck Outline: Urban Vertical Farming Startup**

1. **Problem**
- Urban areas face challenges in consistent fresh food supply due to supply chain disruptions and limited arable land.
- Traditional farming is inefficient and labor-intensive, with high operational costs and environmental impact.
- Underutilized urban real estate remains untapped despite the rising demand for local, sustainable produce.

2. **Solution**
- Modular, scalable vertical farms designed to fit diverse urban spaces and adapt quickly to shifting supply chain conditions.
- Integration of AI and IoT systems for precision crop monitoring and real-time yield optimization.
- Deployment of automation robotics for seeding, harvesting, and labor-intensive tasks to improve cost efficiency.
- Powered by renewable energy solutions to reduce operational costs and maximize sustainability credentials.
- Strategic partnerships with municipal governments to activate underutilized urban real estate and leverage sustainability incentives.

3. **Market**
- Growing urban populations demanding locally sourced, fresh produce.
- Increasing municipal initiatives and funding driving smart city and sustainable urban development projects.
- Expanding investor interest in AgTech and urban farming innovation, especially those emphasizing tech integration and scalability.

4. **Business Model**
- Revenue generated from sales of fresh produce to urban retailers, restaurants, and direct consumers through subscription models.
- Licensing modular farm technology and software platforms to other urban farmers and municipalities.
- Partnering with municipalities on lease or revenue-sharing models to utilize underutilized urban real estate.
- Offering precision farming data services powered by AI and IoT as a subscription SaaS model.

5. **Funding Ask**
- Seeking investment/grants totaling [$X million] to accelerate:
  * R&D in AI/IoT precision monitoring and automation robotics (targeting Automation and Robotics Innovation Grants and AgTech Innovation Grants).
  * Deployment of renewable energy infrastructure in farms (leveraging Sustainability and Renewable Energy Funding Programs).
  * Expansion of partnerships with municipal governments to secure urban sites (aligned with Smart Cities and Urban Development Grants).
  * Scaling modular farming units to enable rapid urban integration (supported by AgTech-Focused Venture Capital Funds).

6. **Team**
- Experienced founders with backgrounds in urban agriculture technology, AI/IoT systems, and sustainable energy solutions.
- Engineering and robotics experts driving automation development.
- Urban planning and government liaison specialists focused on municipal partnerships and smart city initiatives.
- Advisors from AgTech venture capital and sustainability funding bodies.

7. **Traction**
- Prototype modular vertical farm successfully integrated with AI/IoT monitoring showing XX% yield improvement.
- Pilot partnerships established with [City/Municipality Names] to activate underutilized urban spaces.
- Renewable energy systems integrated, reducing energy costs by XX%.
- Initial deployment of automation robotics reducing labor costs by YX%.
- Secured preliminary letters of intent from local retailers and B2B subscription customers.

[INFO] Trace log written to D:\Python_Projects\VT-Projects\10-Dev_Tools_& Product_Readiness\logs\run_20260720T161812-dcb9c2f2.jsonl
(.venv) PS D:\Python_Projects\VT-Projects\10-Dev_Tools_& Product_Readiness>
```

Figure 14 -- CLI Run: urban vertical farming (Trace Log)

The corresponding trace log, confirming the same one-pass behavior as the fintech run.

```
run_20260720T161012-dcb9c2f2.ju x +
File Edit View

{"timestamp": "2026-07-20T16:10:17.642043+00:00", "run_id": "20260720T161012-dcb9c2f2", "agent": "research_agent", "revision_count": 0, "state_read": {"domain": "urban vertical farming", "feedback_notes": ""}, "state_written": {"research_findings": "- Rising investor preference for modular, scalable vertical farms that can quickly adapt to urban spaces and supply chain disruptions. \n- Integration of AI and IoT for precision crop monitoring enabling real-time optimization and yield improvement. \n- Expansion of partnerships between vertical farms and municipal governments to activate underutilized urban real estate with sustainability incentives. \n- Increasing use of renewable energy solutions to lower operational costs and enhance environmental impact credentials. \n- Advances in automation robotics streamlining labor-intensive tasks like seeding and harvesting to improve cost efficiency."}}
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Figure 15 -- Notebook Run: telehealth for elder care

interface/notebook_gui.ipynb run via VS Code's Jupyter extension, proving the second interface produces consistent output through the same workflow.graph.run_workflow() entry point. Note: ipywidgets.Output() content is live kernel state and is not reliably persisted to the .ipynb file on save -- this screenshot, not the saved notebook file, is the record of this run.

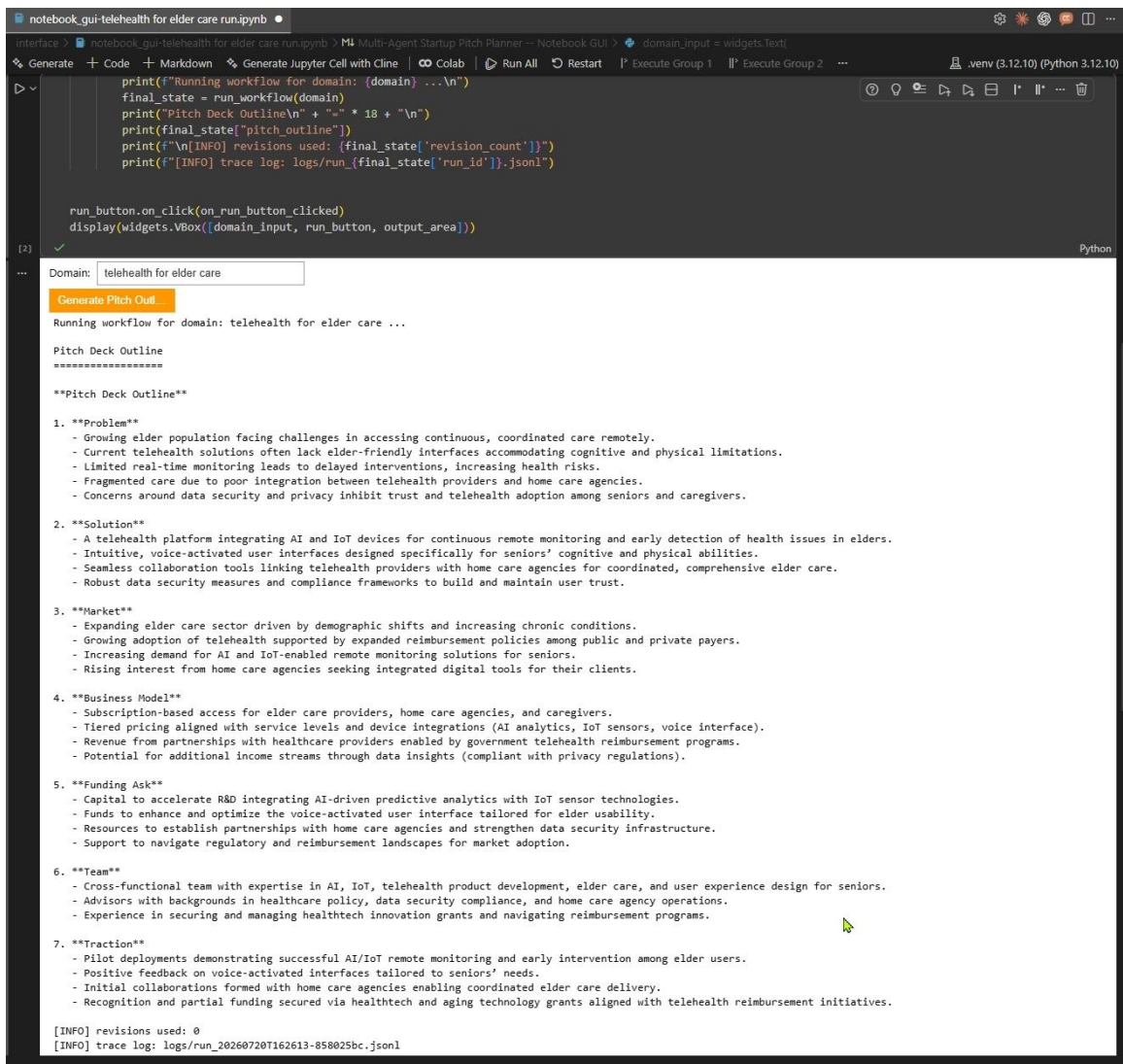


Figure 16 -- Application Insights Telemetry Confirmed

Application Insights Transaction search / end-to-end transaction detail view showing agent.research_agent, agent.funding_advisor, and agent.pitch_coach spans across multiple timestamps, with custom properties (run_id, domain, revision_count, next_step) matching the local trace logs exactly -- confirming the Azure telemetry path works end to end, not just gracefully degrading.

Microsoft Azure

Search resources, services, and docs (G+I)

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appi-accelerator-agents | Search

End-to-end transaction details

Search results

Filtered on: timestamp > 7/20/2026, 74...

timestamp < 7/20/2026, 11...

7/20, 11:34:40 AM - DEPENDENCY
agent_pitch_coach
Call status: True

7/20, 11:34:40 AM - DEPENDENCY
agent_funding_advisor
Call status: True

7/20, 11:26:33 AM - DEPENDENCY
GET /metadata/instance/compute
Base name: 169.254.169.254
Call status: False

7/20, 11:26:30 AM - DEPENDENCY
agent_pitch_coach
Call status: True

7/20, 11:26:24 AM - DEPENDENCY
agent_funding_advisor
Call status: True

7/20, 11:26:18 AM - DEPENDENCY
agent_research_agent
Call status: True

7/20, 11:25:19 AM - DEPENDENCY
agent_pitch_coach
Call status: True

7/20, 11:25:19 AM - DEPENDENCY
GET /metadata/instance/compute
Base name: 169.254.169.254
Call status: False

7/20, 11:25:11 AM - DEPENDENCY
agent_funding_advisor
Call status: True

7/20, 11:10:28 AM - DEPENDENCY
agent_pitch_coach
Call status: True

7/20, 11:10:21 AM - DEPENDENCY
agent_funding_advisor

End-to-end transaction
Operation ID: 7d529acadd4299c7639006cdcd8f6b8

Internal

EVENT	RES	DURATION
agent_pitch_coach	0	0

Graph (Preview)

Traces and events 0 Traces 0 Events

View all

Create work item

agent_pitch_coach

Internal Properties

Show all

Event time	7/20/2026, 11:34:40.632058 AM (Local time)
Type	HttpRequest
Result code	0
Call status	true
Duration	0
Name	agent_pitch_coach

Custom Properties

run_id	20260720T163401-093003c5
agent	pitch_coach
domain	finetech
revision_count	0
next_step	end

Command

Copy

agent_pitch_coach

Related Items

Show what happened before and after this dependency in User Flows